

Syllabus for

Academic year 2021/2022 >

TIF083 - Experimental physics 1 - measuring technique

Experimentell fysik 1 - mätteknik

Syllabus adopted 2021-01-15 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

9,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Engineering Physics

Department: 16 - PHYSICS

Teaching language: Swedish

Application code: 57130

Open for exchange students: No

Maximum participants: 120

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0119 Laboratory 1,5 c Grading: UG	1,5 c						
0219 Project 1,5 c Grading: TH	1,5 c						
0419 Laboratory 1,5 c Grading: UG		1,5 c					
0519 Field trip 0,0 c Grading: UG			0,0 c				
0619 Project 4,5 c Grading: TH			3,0 c	1,5 c			

In programs

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

[Lars Hellberg](#)

[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Course C requires course B.

Aim

Part A, B and C:

To provide training in experimental work and experimental problem solving through laboratory work and projects.

Part D:

Through visits to one or more large-scale research facilities, the students should obtain an overview of: i) the research carried out, ii) the tools used and iii) how to get access to the research facilities.

Learning outcomes (after completion of the course the student should be able to)

Part A:

- show experimental skill
- perform good experimental physics, especially in mechanics
- formulate problems themselves

- design and build experimental setup
- make accurate measurements with advanced equipment
- write technical reports

Part B:

- apply and solve problems regarding basic electrical measurement technology
- build and characterize simpler electrical circuits containing resistors, capacitors and inductors.
- use common instruments for DC and AC measurements
- characterize the properties of diodes, transistors and operational amplifiers as well as build circuits with these components
- read and use component datasheets
- document and analyze measurement results

Part C:

- solve problems regarding advanced electrical measurement technology
- build and troubleshoot experiment settings
- read and use manuals for advanced electrical measuring instruments
- use advanced electrical measuring instruments to independently measure reality related metrological problems in experimental physics
- collect and analyze measurement data and draw conclusions from these
- automate measurements using measuring instruments that are connected to computers and the Labview programming environment
- keep laboratory log notes
- communicate ideas and results by writing preliminary studies and technical reports as well as oral and poster presentations

Part D:

- describe a current research area, the scientific issues addressed and the methods that the current facility can offer.
- describe the underlying technology and the mode of operation of the plant, including the specific pros and cons of these systems.
- provide an overview of the scientific and engineering skills required by a modern research facility.
- describe the operating conditions of the plant, how the selection process for using its resources is done and how it is controlled.

Content

Part A Experimental physics - Mechanics

Part B Basic electrical measurement technology and experimental electronics

Part C Advanced electrical measurement technology

Part D Field trip

Organisation

Pre-studies, Laboratory work, Oral or Poster presentation, Written reports and Field trips

Literature

Laboratory instructions

Examination including compulsory elements

Part A: The assessment is based on a written pre study and a written final report.

Part B: Approved participation in the laboratory exercises.

Part C: For three of the four sub-areas of Part C, the assessment is based on *i.* a written pre study; *ii.* the practical laboratory work in the lab, and *iii.* a written (or in one sub-area oral) final report per sub-area. For the fourth sub-area, which is presented orally to a student group, only the pre study is assessed.

Part D: Approved participation in the study trip

The total grade on the course, given that Part B and Part D are approved, is based on 40% from the result of Part A and 60% from the result of Part C.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.